1.2 Coastal Systems & Landscapes

1. Coasts as Systems

Coastal system: An open system where energy and material cross the boundary. The coast is shaped by the interaction of marine processes (waves, tides, currents), sub-aerial processes (weathering, mass movement), terrestrial processes (river sediment supply), and human activity. The coast exists at the interface of land, sea, and atmosphere.

Key system components:

- **Inputs:** wave energy, tidal energy, wind energy, sediment from rivers/cliff erosion/offshore, solar energy, precipitation, geological structure
- **Outputs:** sediment removed by longshore drift or offshore currents, erosion products, evaporation, dissolved minerals
- **Stores:** beaches, sand dunes, spits, bars, tombolos, mudflats, salt marshes, cliff faces
- **Transfers:** longshore drift, onshore/offshore sediment movement, aeolian (wind) transport, mass movement
- **Feedback:** negative (cliff collapse adds sediment to beach, wider beach absorbs wave energy, reducing further erosion) and positive (sea-level rise narrows beach, less wave absorption, more erosion, beach narrows further)

Sediment cell (littoral cell): A largely self-contained stretch of coastline within which sediment movement occurs in a cycle of sources, transfers, and sinks. England and Wales have 11 major sediment cells (e.g. Cell 1: St Abb's Head to Flamborough Head; Cell 2: Flamborough Head to the Humber). Sediment rarely crosses cell boundaries, making them useful units for coastal management and Shoreline Management Plans.

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Sediment budget: The balance between sediment inputs (sources) and outputs (sinks) within a sediment cell. If inputs exceed outputs, the coast builds out (progradation). If outputs exceed inputs, the coast erodes (retrogradation). Human interference (e.g. groynes trapping sediment, dam construction reducing river sediment) can disrupt the budget.

2. Waves

Wave: An undulation in the water surface caused by wind friction transferring energy to the water. In deep water, water particles move in circular orbits (no net forward movement of water). The size of a wave depends on *wind speed*, *wind duration*, and *fetch* (the distance of open water over which the wind blows). The UK's prevailing south-westerly winds produce a dominant fetch across the Atlantic (~5,000 km).

Wave Characteristics

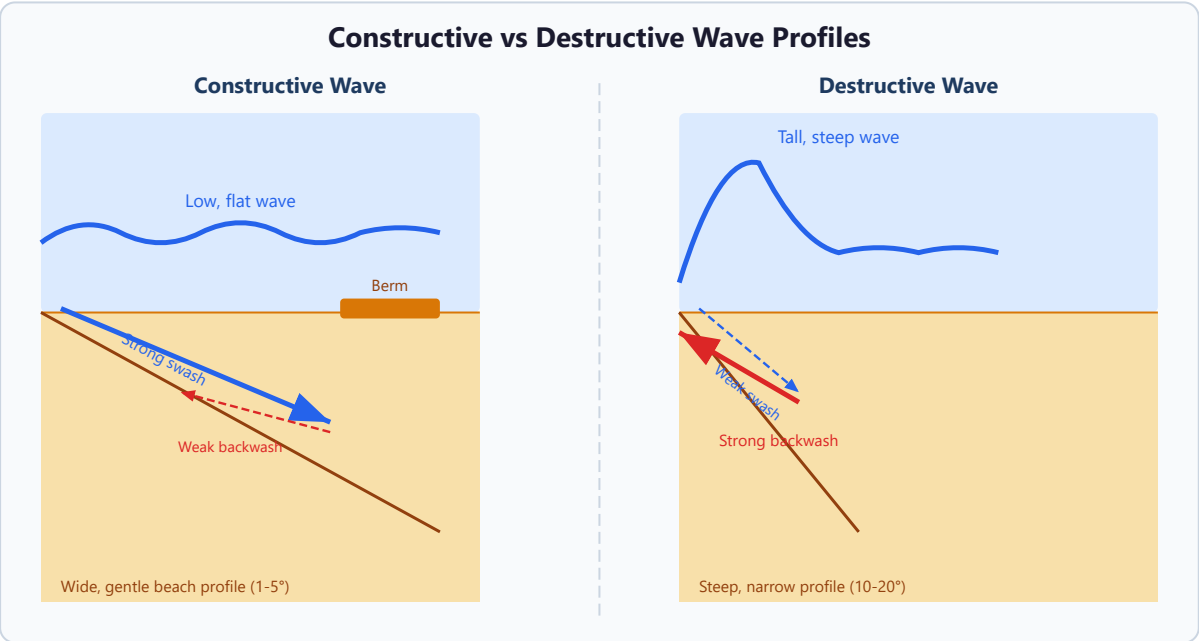
- **Wave height** — vertical distance between trough and crest
- **Wave length** — horizontal distance between successive crests
- **Wave frequency** — number of waves per minute
- **Wave period** — time between successive wave crests passing a fixed point (typically 6–14 seconds)
- **Wave steepness** — ratio of wave height to wave length. When steepness exceeds 1:7, the wave becomes unstable and breaks.
- **Wave energy** — proportional to the square of wave height. Doubling wave height quadruples energy.

Wave breaking: As waves approach the shore, friction with the seabed slows the base of the wave. The top continues at the original speed, causing the wave to steepen and eventually break. Three types of breaking wave: *spilling* (gentle slope, wave gradually spills forward), *plunging* (steeper slope, wave curls and crashes), and *surging* (very steep slope, wave surges up beach without breaking).

Constructive vs Destructive Waves

Feature	Constructive Waves	Destructive Waves

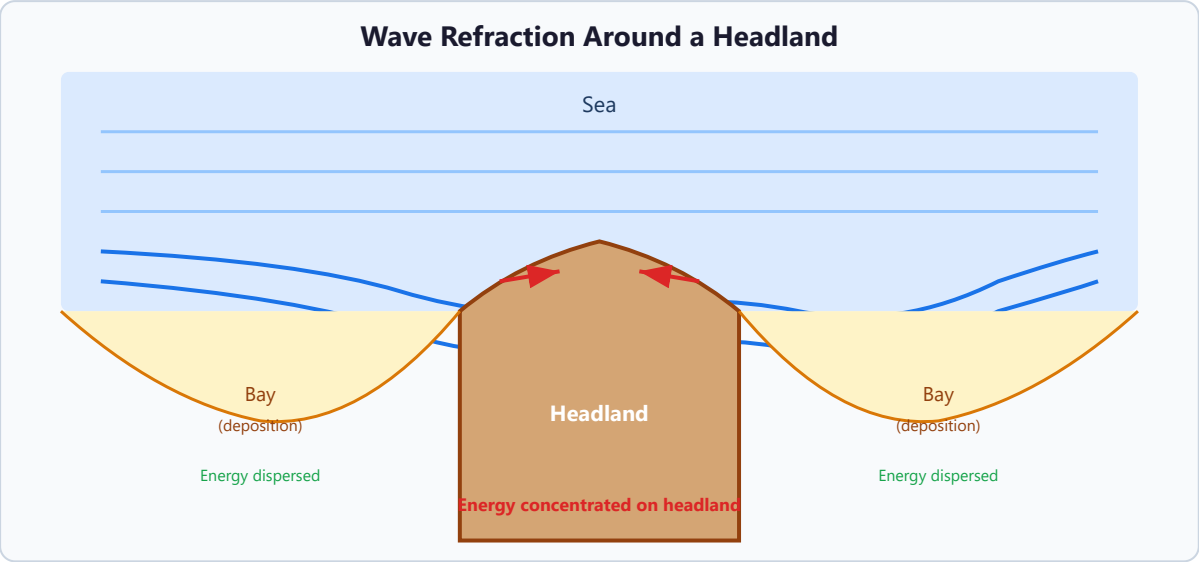
Frequency	Low (6–8 per minute)	High (10–14 per minute)
Wave height	Low (<1 m typically)	High (>1 m, often 3–5 m in storms)
Wavelength	Long	Short
Swash vs backwash	Strong swash, weak backwash	Weak swash, strong backwash
Effect on beach	Deposits sediment, builds up beach material	Removes beach material, erosion dominates
Beach profile	Wide, gently sloping, with berms	Steep, narrow, with scoured surface
Typical conditions	Calm weather, long swell waves from distant storms	Storm conditions, locally generated short-period waves
Breaking type	Spilling breakers	Plunging breakers



Wave Refraction

When waves approach an irregular coastline with headlands and bays, they bend (refract) due to differential friction with the seabed. Waves slow in shallow water near headlands, causing wave fronts to curve and concentrate energy on the headland (explaining why headlands erode). In bays, waves spread

out and energy is dissipated over a wider area, encouraging deposition and beach formation. Over geological time, wave refraction tends to straighten irregular coastlines.



3. Coastal Processes

Marine Erosion

- **Hydraulic action** — the force of water and trapped air being compressed into cracks and joints in the rock. Pressures can reach 30 tonnes/m^2 during storms, widening fractures and eventually dislodging blocks of rock.
- **Abrasion (corrasion)** — rock fragments carried by waves act like sandpaper, scraping and wearing away the cliff face. Most effective during storms when large pebbles and cobbles are hurled at cliffs. The dominant erosion process on rocky coasts.
- **Attrition** — rock fragments carried by the sea collide with each other, becoming progressively smaller, smoother, and more rounded. Angular boulders become rounded pebbles and eventually sand. This explains why sediment becomes finer with distance along a beach.
- **Corrosion (solution)** — weak carbonic acid in seawater (formed from dissolved CO_2) dissolves soluble rocks like limestone ($CaCO_3$) and chalk. Particularly effective in warm, slightly acidic water. Creates features like blowholes and narrow inlets on limestone coasts.

Sub-aerial Processes

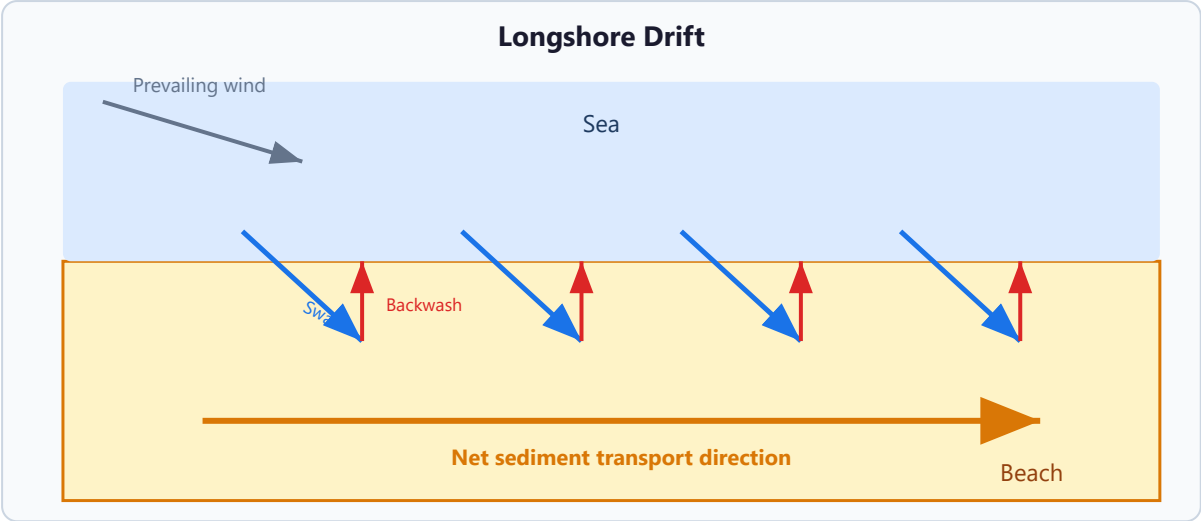
- **Weathering** — the in-situ breakdown of rock without movement:

- *Mechanical (physical):* freeze-thaw (water expands by 9% when frozen, exerting up to 2,100 kg/cm^2 pressure in cracks), salt crystallisation (salt spray dries in cracks, crystals grow and exert pressure), wetting and drying (clay minerals expand when wet and contract when dry, weakening rock)
- *Chemical:* carbonation ($CO_2 + H_2O \rightarrow H_2CO_3$, which dissolves $CaCO_3$ in limestone/chalk), oxidation (iron-bearing minerals rust and weaken), hydrolysis (water reacts with minerals to form clay)
- *Biological:* plant roots widen cracks, burrowing organisms (piddock molluscs bore into soft rock), organic acids from lichens and algae dissolve rock surfaces

- **Mass movement** — downslope movement of material under gravity, triggered by undercutting of cliffs by wave erosion, saturation by rainfall, or loading:
  - *Rockfall:* sudden collapse of individual rocks or blocks from a steep/overhanging cliff face. Common on well-jointed hard rock cliffs (e.g. chalk, limestone).
  - *Landslide:* a block of material slides along a planar (flat) failure surface. Common where a permeable rock overlies impermeable rock (e.g. chalk over clay).
  - *Rotational slump:* material slides along a curved, concave failure surface, rotating backwards. Common in clay cliffs (e.g. London Clay cliffs of the Isle of Sheppey). Often triggered by prolonged rainfall saturating clay.
  - *Mudflow/earthflow:* saturated soil and regolith flows downslope as a viscous mass. Fastest type of mass movement.
  - *Soil creep:* very slow downhill movement of soil particles, causing tilted fences and curved tree trunks.

Sediment Transport

Longshore drift (littoral drift): The movement of sediment along the coast by wave action. Waves approach the beach at an angle (determined by the prevailing wind direction), carrying sediment diagonally up the beach with the swash. Backwash carries sediment straight back down the beach under gravity. The net result is a zig-zag movement of material along the shore in the direction of the prevailing wind. In England, the dominant drift direction is west to east along the south coast and south to north along the east coast.



Sediment is transported by waves and currents through four mechanisms:

- **Traction** — large particles (boulders, cobbles) rolled along the seabed by strong waves
- **Saltation** — pebbles and gravel bounced along the seabed in a hopping motion
- **Suspension** — fine particles (silt, clay) carried within the water column
- **Solution** — dissolved minerals carried invisibly in the water (e.g. calcium carbonate from limestone erosion)

4. Erosional Landforms

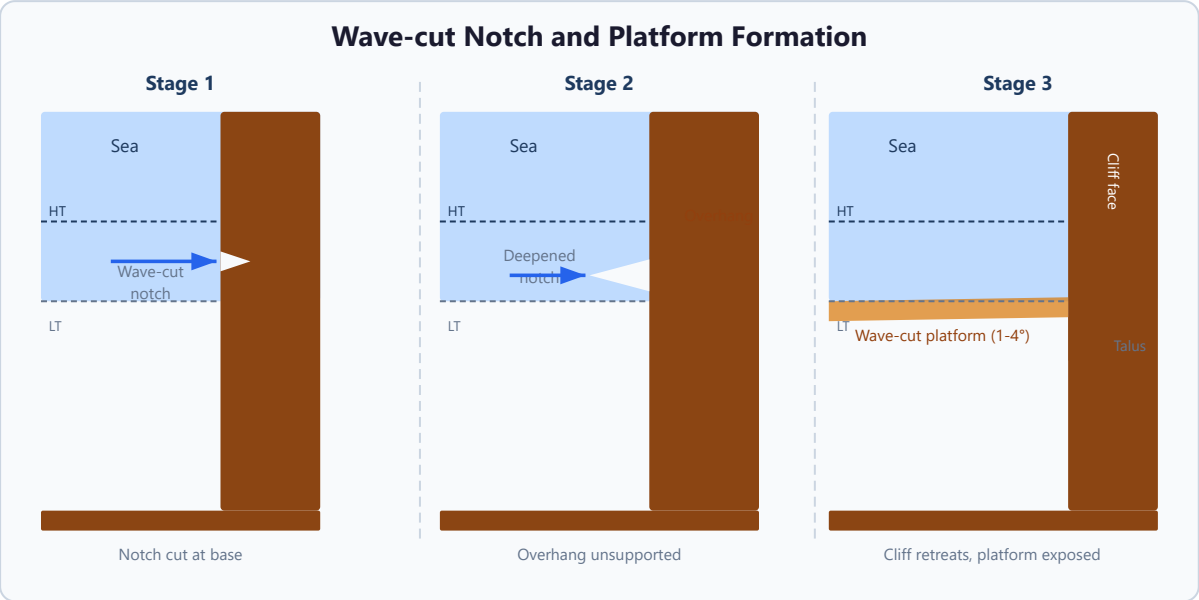
Headlands and Bays

Form along coastlines with alternating bands of resistant (hard) and less resistant (soft) rock perpendicular to the coast (discordant coastline). Differential erosion removes softer rock (e.g. clay, sand) faster, creating bays, while harder rock (e.g. chalk, limestone, granite) resists erosion and remains as protruding headlands. Wave refraction then concentrates energy on headlands. On concordant coastlines (rock bands parallel to coast), a breach in the resistant outer rock can allow rapid erosion of softer rock behind, forming a cove (e.g. Lulworth Cove, Dorset).

Cliff Recession and Wave-cut Platforms

Waves attack the base of a cliff between the high and low tide marks, eroding a **wave-cut notch** through hydraulic action and abrasion. As the notch deepens, the overhanging cliff becomes unsupported and collapses under gravity. Repeated cycles of notch erosion and cliff collapse cause the cliff to **retreat** landward, leaving a gently sloping (typically 1–4°) **wave-cut platform** at the base, exposed at low tide. The

platform eventually limits further erosion because waves break further from the cliff as the platform widens, dissipating energy. Average cliff recession rates in the UK: chalk cliffs 0.1–1 m/yr, soft clay cliffs up to 2–10 m/yr (e.g. Holderness, up to 2 m/yr).



Caves, Arches, Stacks, and Stumps

A sequence of erosional landforms developing on headlands:

1. **Cave:** Waves exploit a joint, fault, or area of weakness in the headland through hydraulic action and abrasion, eroding a cave (hollow) into the rock face.
2. **Arch:** When caves on opposite sides of a narrow headland break through and meet, they form a natural arch. The roof is weathered from above and eroded from below.
3. **Stack:** Continued erosion and weathering cause the roof of the arch to collapse, leaving an isolated pillar of rock — a stack (e.g. Old Harry Rocks, Dorset; The Needles, Isle of Wight).
4. **Stump:** The stack is undercut at the base, eventually collapsing to leave a low stump, often only visible at low tide.

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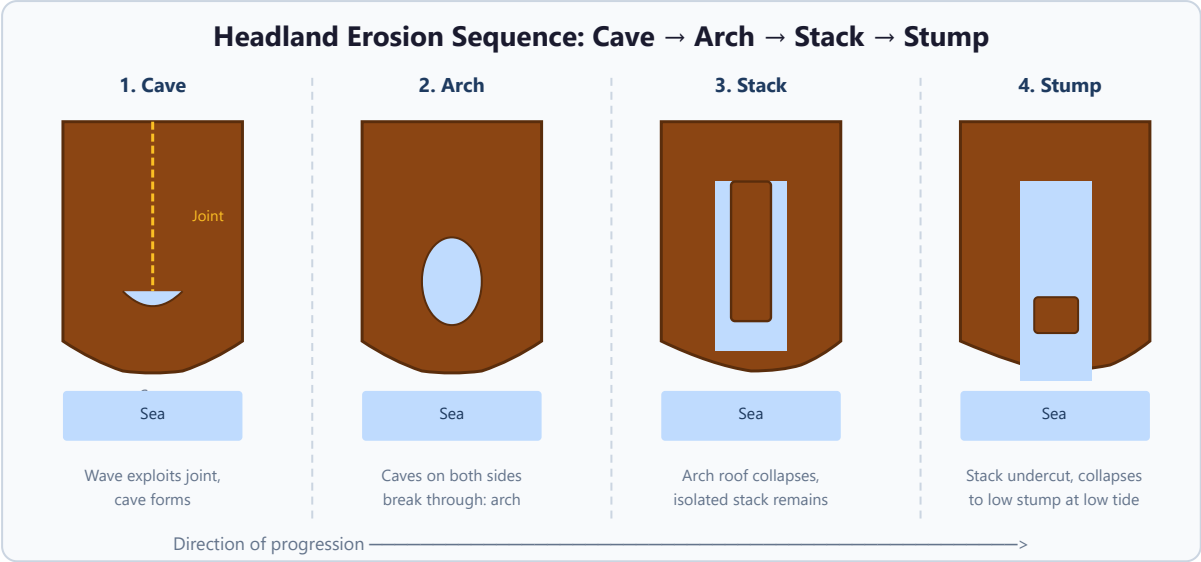


Geo and Blowhole

A **geo** is a narrow inlet formed when the roof of a cave collapses inland, or when waves erode along a fault line. A **blowhole** forms when a vertical crack above a cave is widened by hydraulic pressure from waves entering the cave below; spray is forced upward through the hole during storms.

Case Study: The Dorset Coast (Jurassic Coast World Heritage Site)

The Dorset coast stretches 155 km from Exmouth to Studland and displays a remarkable range of erosional and depositional landforms due to its diverse geology. **Lulworth Cove** is a classic concordant coastline feature: waves breached the resistant Portland Limestone outer band and rapidly eroded the softer Wealden Clay behind, creating a near-circular cove 450 m wide. **Durdle Door** is a natural limestone arch formed by marine erosion exploiting weaknesses in the Portland Limestone headland. **Old Harry Rocks** (Handfast Point) are chalk stacks and stumps formed by erosion of the chalk headland at the eastern end of the Jurassic Coast. The coast demonstrates the entire sequence from cave to stump. Recession rates on the chalk cliffs here average 0.1–0.5 m/yr. The Jurassic Coast was designated a UNESCO World Heritage Site in 2001 for its geological significance, covering 185 million years of Earth history.

5. Depositional Landforms

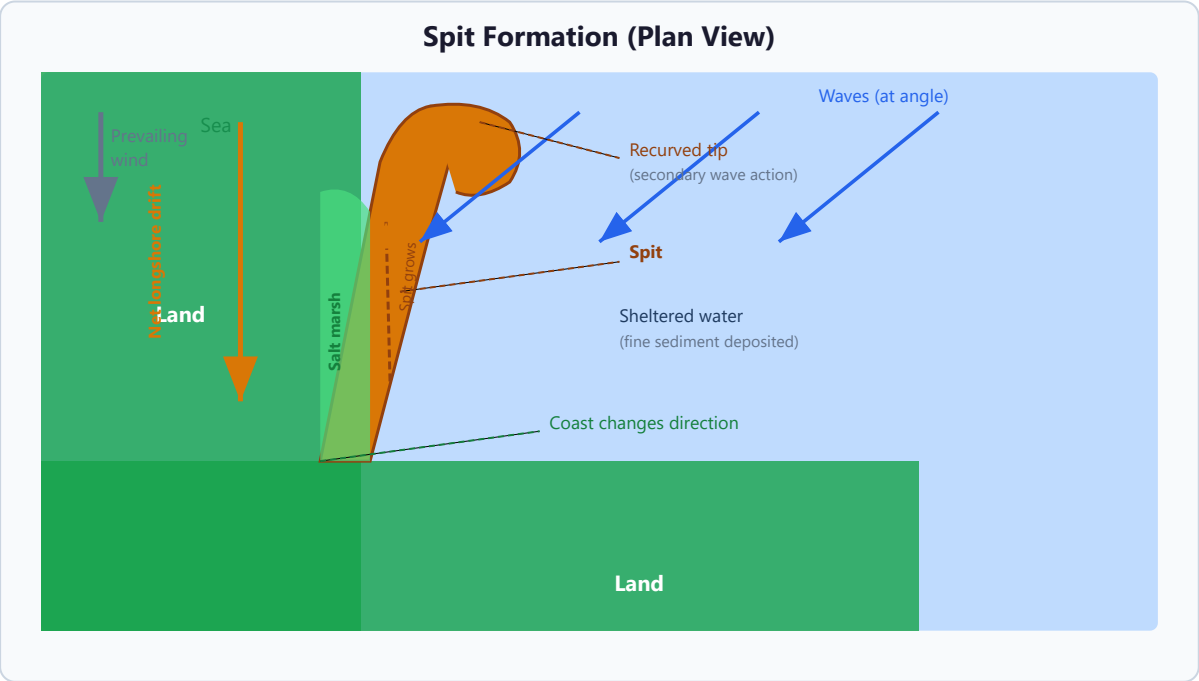
Beaches

Beaches are accumulations of sand, shingle, or pebbles deposited by constructive wave action and longshore drift. They form wherever the rate of sediment supply exceeds the rate of removal.

- **Sandy beaches** — wide, gently sloping (typically 1–5°), formed by fine material. Sand particles are small enough to pack together, creating a smooth surface. Common in sheltered bays.
- **Shingle/pebble beaches** — steeper (10–20°) because larger particles are more permeable (water drains through, weakening backwash). Often found on exposed coasts. Composed of flint, limestone, or other resistant rock fragments.
- **Beach features:** berms (ridges deposited at high tide marks), cusps (crescent-shaped hollows formed by swash-backwash interaction), runnels (shallow channels running parallel to the shore).
- **Beach profile** — changes seasonally: summer profiles are wider with gentle gradients (constructive waves deposit material), winter profiles are narrower and steeper (destructive waves remove material, creating offshore bars).

Spits

A spit is a long, narrow finger of sand or shingle extending from the coastline into the sea, usually at a point where the coast changes direction (e.g. at a river mouth or change in coastline orientation). Formed by longshore drift continuing to transport sediment beyond the change in coastline direction. The tip may curve due to secondary wave action from a different direction. Sheltered water behind the spit often develops into mudflats and salt marsh, created by deposition of fine sediment in calm conditions.



Case Study: Spurn Head, East Yorkshire

Spurn Head is a 5.5 km long sand and shingle spit extending across the mouth of the Humber Estuary. It has formed through southward longshore drift from the rapidly eroding Holderness coast (which loses ~2 m/yr of boulder clay cliffs, providing abundant sediment). The spit has a recurved tip (hooked end) formed by wave refraction from the estuary side. Behind the spit, extensive mudflats and salt marshes have developed in the sheltered Humber Estuary. Spurn Head is naturally dynamic — historical records show it has been breached and rebuilt multiple times (roughly every 250 years). In December 2013, a major tidal surge breached the spit, severing the road and making it accessible only at low tide. The RNLI lifeboat station was relocated. Spurn is now managed by the Yorkshire Wildlife Trust as a "managed realignment" site, accepting natural processes rather than defending the spit.

Bars and Tombolos

- **Bar:** Forms when a spit grows across a bay, completely enclosing it and creating a lagoon behind (e.g. Slapton Ley, Devon — a freshwater lagoon behind a shingle bar). Bars form in bays with relatively low-energy wave environments.
- **Tombolo:** A bar connecting an island to the mainland (e.g. Chesil Beach connecting the Isle of Portland to the Dorset coast, 28 km long, with sediment graded from pea-sized pebbles in the west to cobble-sized in the east).
- **Offshore bar (breakpoint bar):** A submerged ridge of sand running parallel to the shore, formed where destructive waves deposit sediment at their break point.

Sand Dunes

Sand dunes form where prevailing onshore winds blow dry sand inland from a wide, sandy beach. Obstacles (driftwood, vegetation) trap sand, which accumulates. Key features:

- **Embryo dunes** — small, newly formed dunes nearest the beach. Pioneer species (e.g. sea rocket, sand couch grass) colonise. Highly mobile, 1–2 m height.
- **Fore dunes (yellow dunes)** — 5–10 m high, colonised by marram grass (Ammophila arenaria) with deep roots that bind sand. Still largely sand with little humus. pH ~8 (alkaline, due to shell fragments).
- **Grey dunes** — further inland, more stable, with greater plant diversity. Named for their grey colour from increasing humus content. pH decreasing (~6–7). Mosses, lichens, and wildflowers.

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- **Dune slacks** — wet depressions between dunes where the water table is near or at the surface. Support wetland plant communities (e.g. creeping willow, orchids).
- **Climax vegetation** — furthest inland, mature woodland or heathland. Soil is well-developed with thick humus. pH ~4–5 (acidic). This ecological succession from pioneer to climax is called a **psammosere**.

Salt Marshes

Salt marshes develop in sheltered, low-energy environments (behind spits, in estuaries) where fine sediment (silt and mud) is deposited by tides. Pioneer species (glasswort/Salicornia, cord grass/Spartina) colonise the mudflats, trapping more sediment and raising the surface. Over time, a succession (**halosere**) develops from pioneer salt-tolerant species to upper marsh communities (sea lavender, sea aster, rushes). Salt marshes are ecologically valuable: nurseries for fish, habitats for wading birds, carbon sinks (sequestering carbon at rates 2–4 times greater than tropical forests per hectare), and natural flood defences.

6. Sea Level Change

Eustatic change: A global change in sea level caused by a change in the volume of water in the oceans (e.g. melting/formation of ice sheets) or a change in the shape of ocean basins. During the last glacial maximum (~20,000 years ago), sea level was ~120 m lower than today.

Isostatic change: A local or regional change in the relative level of land and sea caused by the land rising or falling. During glaciation, the weight of ice depresses the land; after deglaciation, the land slowly rebounds (isostatic uplift). Scotland is currently rising at ~1–3 mm/yr, while SE England is sinking at ~1–2 mm/yr.

Emergent Coastlines (land rising relative to sea)

- **Raised beaches** — former beaches now elevated above current sea level, sometimes with old cliff lines, wave-cut platforms, and caves behind them (e.g. raised beaches in western Scotland at 8–10 m above present sea level).
- **Relict cliffs** — fossil cliff lines no longer subject to marine erosion, often vegetated.

Submergent Coastlines (sea rising relative to land)

- **Ria** — a drowned river valley with a progressively deeper and wider mouth seaward. Formed by post-glacial sea-level rise flooding existing river valleys (e.g. Kingsbridge Estuary, Devon; Milford Haven, Wales). Characterised by a branching, irregular shape following the original valley.
- **Fjord** — a drowned glacial valley with steep, near-vertical sides, a U-shaped cross-section, and a shallow threshold (terminal moraine) at the mouth (e.g. Sognefjorden, Norway: 204 km long, 1,308 m deep). Formed when post-glacial sea-level rise flooded deep glacial troughs.
- **Dalmatian coast** — sea-level rise floods valleys running parallel to the coast, creating elongated islands and channels (e.g. the Croatian coast).

Current and Future Sea Level Rise

Global mean sea level has risen ~20 cm since 1900 and is currently rising at ~3.6 mm/yr (accelerating). Causes: thermal expansion of ocean water (~40%), melting of glaciers and ice caps (~30%), and melting of Greenland and Antarctic ice sheets (~30%). IPCC projections: 0.43–0.84 m rise by 2100 under moderate scenarios; up to 1.1+ m under high-emission scenarios. Consequences: increased coastal flooding, saltwater intrusion into freshwater aquifers, erosion, loss of low-lying land, and threats to coastal settlements and ecosystems.

7. Coastal Management

Shoreline Management Plan (SMP): A large-scale strategic plan covering an entire sediment cell, developed by coastal authorities, setting out management policy for each stretch of coast. Four policy options: Hold the Line, Advance the Line, Managed Realignment, and No Active Intervention.

Hard Engineering

Method	How It Works	Advantages	Disadvantages	Cost (approx.)
Sea wall	Concrete/stone wall at cliff base reflects wave energy	Effective protection, long lifespan (50+ yrs), public confidence	Very expensive, visually intrusive, reflected wave energy scours beach at base, can create wave overtopping	£5,000–10,000/m

Groynes	Wooden or rock barriers perpendicular to shore, trapping longshore drift	Builds up beach (natural defence), relatively inexpensive	Starves downdrift areas of sediment, causing accelerated erosion elsewhere (terminal groyne syndrome)	£5,000–10,000 each
Rock armour (rip-rap)	Large boulders placed at cliff base to absorb wave energy	Effective, relatively cheap, absorbs rather than reflects energy	Can be visually unattractive, rocks shift in storms, not always accessible	£1,000–3,000/m
Gabions	Wire cages filled with rocks placed at cliff base	Cheap, absorb wave energy, permeable	Unattractive, short lifespan (5–10 yrs), wire corrodes	£100–500/m
Offshore breakwaters	Rock or concrete barriers placed offshore, parallel to coast	Reduce wave energy before it reaches shore, encourage deposition behind	Expensive, interfere with navigation, can cause unexpected sediment movement	£5,000+/m
Revetments	Sloped structures on cliff face that absorb/dissipate wave energy	Effective, less expensive than sea walls	Can be undermined, limited lifespan	£2,000–5,000/m

Soft Engineering

Method	How It Works	Advantages	Disadvantages
Beach nourishment	Adding sand/shingle from elsewhere to widen and raise the beach	Natural-looking, absorbs wave energy, recreational value	Needs regular replenishment (every 5–10 yrs), expensive over time (£3,000–5,000/m), may use dredged material of different grain size

Managed realignment	Allowing the sea to flood low-value areas, creating new salt marsh and mudflats	Creates habitats, provides natural flood defence, cheaper than hard engineering, sustainable	Loss of farmland, compensation costs, local opposition, slow to establish
Dune stabilisation	Planting marram grass, installing sand fencing, restricting access	Natural, cheap, enhances biodiversity	Slow, requires management of tourism, vulnerable to storms
Cliff drainage	Installing drains to reduce water saturation in cliff, reducing mass movement risk	Reduces slumping and landslides	Requires maintenance, does not prevent marine erosion
Beach recycling	Moving sediment from where it accumulates (downdrift end) back to where it is needed (updrift)	Low cost, maintains natural beach	Temporary, requires regular intervention

Case Study: Holderness Coast, East Yorkshire — Rapid Erosion

The Holderness coast is the fastest-eroding coastline in Europe, retreating at an average of 1.8 m/yr (up to 10 m in some storms). It stretches 61 km from Flamborough Head (resistant chalk) to Spurn Head. The cliffs are composed of glacial till (boulder clay) — a soft, easily eroded material. Since Roman times, ~4 km of land has been lost, and over 30 villages have disappeared (e.g. Ravenser Odd, a medieval port). Key factors: weak geology, powerful North Sea waves (long fetch from the NNE), limited beach material for natural protection (the clay produces little sand/gravel), and rising sea levels. Management is controversial: Mappleton received £2 million of rock armour groynes in 1991, which protected the village but starved beaches to the south, accelerating erosion at Great Cowden (which now has no protection). Bridlington has a sea wall; Withernsea has groynes and a sea wall. Spurn Head is left undefended ("No Active Intervention"). The Holderness coast illustrates the conflict between protecting some areas and accepting loss elsewhere — a key coastal management dilemma.

Case Study: Medmerry, West Sussex — Managed Realignment

In 2013, the Environment Agency completed the UK's largest open-coast managed realignment scheme at Medmerry. A new 7 km earth embankment was built inland, and the old shingle bank was deliberately breached, allowing the sea to flood 183 hectares of low-lying farmland. Cost: £28 million (compared to an estimated £300+ million for maintaining traditional hard defences over 100 years). Benefits: 348 properties now better protected from flooding, creation of intertidal habitat (salt marsh, lagoons) supporting populations of avocets, little terns, and water voles, and carbon sequestration by new salt marsh. The RSPB manages the site as a nature reserve. Medmerry demonstrates how managed realignment can deliver flood protection, environmental benefits, and cost savings simultaneously. Critics note the loss of productive farmland and the disruption to local landowners during construction.

Exam Tip: When evaluating coastal management strategies, always consider: (1) economic costs and benefits, (2) environmental impacts (both positive and negative), (3) social impacts (who benefits, who loses), (4) sustainability (short-term fix vs long-term solution), and (5) how protecting one area may affect adjacent areas (sediment budget effects). Use specific named examples with data.

Exam Tip: AQA expects you to understand the coast as a *system*. Always link processes to landforms to management. For example: longshore drift (process) creates spits (landform), and groynes (management) interrupt longshore drift, causing downdrift erosion. Think about inputs, outputs, stores, and feedback when structuring answers about coastal change.

Exam Tip: In 20-mark essays on coastal management, develop a clear evaluative argument. The best answers consider whether management should focus on protecting people and property (engineering approach) or working with natural processes (managed retreat/realignment). There is no single "right" answer — the approach depends on economic value, environmental significance, and community needs.

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